

Topic: The Impact of Conditional Cash Transfers (SURE-P MCH Program) on Maternal Autonomy in Nigeria

This paper seeks to examine the impact of the Subsidy Reinvestment & Empowerment Program – Maternal and Child Health (SURE-P MCH) conditional cash transfer scheme which was launched in 2012 in Nigeria improved women’s decision-making autonomy within households.

I’ll be using the DHS data from 2008 (pre-program) and 2013 & 2018 (post-program), employing a DiD (difference-in-differences) design comparing between states that implemented the CCT program with those that did not.

Background

Nigeria as a developing country still faces high rates of maternal and child mortality partly linked to the low utilization of antenatal care and health facility services. Generally, cash transfers (CCTs) are used in developing countries to incentivize health service utilization. However, the impact of these CCTs on women’s autonomy which is a deeper social and behavioral outcome, is less studied. This study provides evidence from Nigeria, using a national program to evaluate whether these financial incentives shift household decision-making power.

The Subsidy Reinvestment and Empowerment Program (SURE-P) was introduced in 2012 by the Federal Government of Nigeria following partial fuel subsidy removal. The Maternal and Child Health (MCH) project, included a Conditional Cash Transfer (CCT) scheme where pregnant women received cash transfers conditional on completing a series of health visits such as registering for antenatal care (ANC), attending four ANC visits, delivering in a health facility and for completing postnatal care visits. The objective was to reduce financial barriers, incentivize facility-based delivery, and improve maternal and child health outcomes. The program was supported by federal resources and implemented in partnership with selected states with high maternal mortality and weaker health service uptake. The main implementation states were:

- Anambra
- Bauchi
- Kaduna
- Niger
- Ondo
- Zamfara

This selective rollout creates a natural experiment for studying causal impacts. By comparing changes in autonomy before and after the program in treatment versus control states, I assess whether the SURE-P CCT scheme influenced maternal autonomy outcomes.

Research Design

This study evaluates the causal impact of the SURE-P MCH conditional cash transfer (CCT) program, launched in 2012 in Nigeria, on women's decision-making autonomy using a difference-in-differences (DiD) approach using three survey rounds (2008-pre-program, 2013-short-run post, and 2018-long-run post).

- **Outcome (Y_{ist}):** Autonomy index for women 'i' in state 's' and year 't' gotten from the DHS survey questions:
 - I. Who decides on woman's own healthcare
 - II. Who decides on large household purchases
 - III. Who decides on visits to relatives/friends
- **Treatment: ($Treat_s$):** whether a woman lives in a state where SURE-P was implemented (Anambra, Bauchi, Kaduna, Niger, Ondo, Zamfara).
- **Post periods ($Post_t$):** Dummy variables for the post-program years (2013 and 2018), with 2008 as the baseline.

{Each question is coded 1 if the woman decides alone or jointly and 0 otherwise. The average of this forms the autonomy}

The treatment effect here will be the change in autonomy in treated states (before vs. after SURE-P) compared to the change in autonomy over the same years in control states.

Potential Outcomes

For each woman, I imagine two possible outcomes:

$Y_{ist}(1)$: her autonomy if her state got the SURE-P program

$Y_{ist}(0)$: her autonomy if her state did not get it

The true effect of the program will be the difference in these:

$$Y_{ist}(1) - Y_{ist}(0)$$

Using DiD solves the missing counterfactual problem by using the control states as a stand-in for what would have happened in the treated states had the program not been rolled out.

The treatment effect in 2013:

$$\beta_{2013} = (\bar{Y}_{treat_{2013}} - \bar{Y}_{treat_{2008}}) - (\bar{Y}_{ctrl_{2013}} - \bar{Y}_{ctrl_{2008}})$$

And in 2018:

$$\beta_{2018} = (\bar{Y}_{treat_{2018}} - \bar{Y}_{treat_{2008}}) - (\bar{Y}_{ctrl_{2018}} - \bar{Y}_{ctrl_{2008}})$$

The DiD regression equation would be:

$$Y_{ist} = \alpha + \beta_{2013}(Treat_s \times Post_{2013}) + \beta_{2018}(Treat_s \times Post_{2018}) + \gamma X_{ist} + \delta_s + \lambda_t + \varepsilon_{ist}$$

X_{ist} are controls for woman's age, education, wealth, etc.

δ_s are state fixed effects (to absorb time variant differences across states.

λ_t are survey year fixed effects

Errors are clustered at state level and DHS sampling weights are used.

The coefficients are;

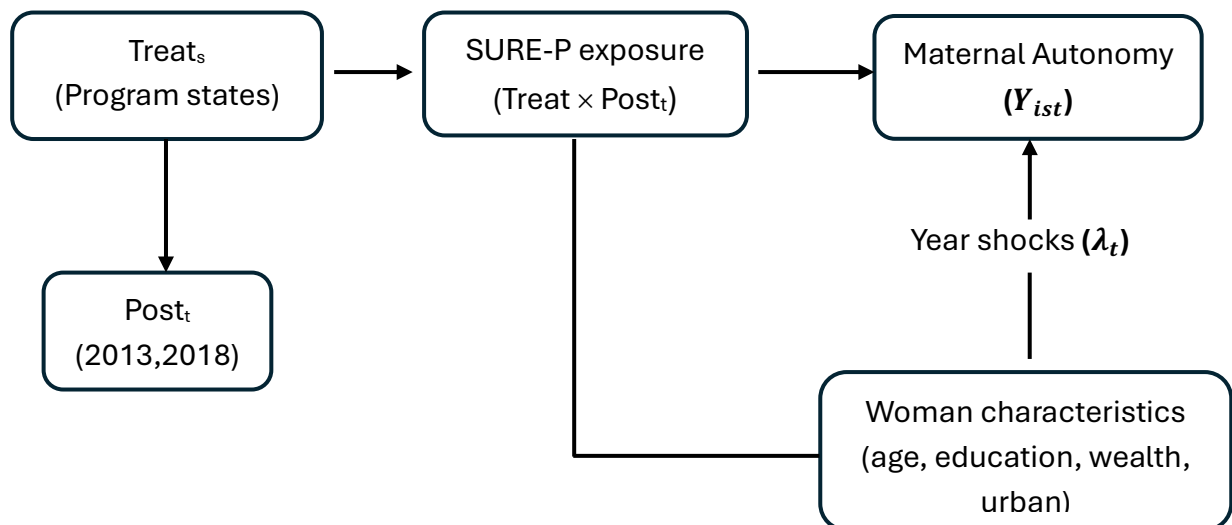
β_{2013} = short-run effect (just after program roll-out)

β_{2018} = long-run effect (to check if changes persisted or grew over the years)

Assumptions

- The key assumption is parallel trends: without SURE-P, autonomy in treated and control states would have moved similarly over time. I would be comparing baseline autonomy and demographics, adding controls and fixed effects, and testing robustness with alternative outcomes and subgroups.
- SUTVA/ No spillovers: Women in non-program states are unaffected by SURE-P rollout in other states.
- No compositional bias: Migration and sample selection do not differentially affect treated states across survey years.
- No anticipation: Autonomy outcomes in 2008 are unaffected by expectations of the 2012 program.

DAG



Data

I pool the Nigeria DHS Individual Recode (IR) files for 2008 (pre), 2013 (short-run post), and 2018 (long-run post). The unit is women 15–49. The main outcome is a maternal autonomy index built from three DHS decision questions (who decides about (i) the woman’s health care, (ii) large household purchases, (iii) visits to relatives/friends, coded 1 if she decides alone or jointly, 0 otherwise, then averaged to 0–1). Treatment is state-level SURE-P participation; I code program states as treated and all others as controls. I control for age, education, wealth, and urban/rural, and use DHS weights with state and year fixed effects in the main analysis.

To be continued...

Variable name	Storage type	Display format	Value label	Variable label
v743a	byte	%8.0g	V743A	person who usually decides on respondent's health care
v743b	byte	%8.0g	V743B	person who usually decides on large household purchases
v743c	byte	%8.0g	V743C	na – person who usually decides on household purchases for daily needs
v743d	byte	%8.0g	V743D	person who usually decides on visits to family or relatives
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. tab region, missing
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region	Freq.	Percent	Cum.
north central	20,389	17.86	17.86
north east	20,486	17.95	35.81
north west	27,099	23.74	59.55
south east	13,700	12.00	71.55
south south	15,951	13.97	85.52
south west	16,529	14.48	100.00
Total	114,154	100.00	

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. tab state, missing
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state of residence	Freq.	Percent	Cum.
abia	2,523	2.21	2.21
adamawa	3,223	2.82	5.03
akwa ibom	2,756	2.41	7.45
anambra	2,795	2.45	9.90
bauchi	3,533	3.09	12.99
bayelsa	2,840	2.49	15.48
benue	3,133	2.74	18.22
borno	3,041	2.66	20.89
cross river	2,248	1.97	22.86
delta	2,708	2.37	25.23
ebonyi	3,349	2.93	28.16
edo	2,583	2.26	30.43
ekiti	2,380	2.08	32.51
enugu	2,711	2.37	34.89
fct-abuja	2,799	2.45	37.34
gombe	3,437	3.01	40.35
imo	2,322	2.03	42.38
jigawa	3,635	3.18	45.57
kaduna	3,934	3.45	49.01
kano	5,448	4.77	53.79
katsina	3,980	3.49	57.27
kebbi	3,485	3.05	60.32
kogi	2,644	2.32	62.64
kwara	2,609	2.29	64.93
lagos	4,179	3.66	68.59
nasarawa	2,948	2.58	71.17
niger	3,283	2.88	74.05
ogun	2,112	1.85	75.90
ondo	2,561	2.24	78.14
osun	2,784	2.44	80.58
oyo	2,513	2.20	82.78
plateau	2,973	2.60	85.38
rivers	2,816	2.47	87.85
sokoto	3,324	2.91	90.76
taraba	3,875	3.39	94.16
yobe	3,377	2.96	97.12
zamfara	3,293	2.88	100.00
Total	114,154	100.00	

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. label var treat "SURE-P state (treated)"
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. tab treat
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SURE-P state (treated)	Freq.	Percent	Cum.
0	94,755	83.01	83.01
1	19,399	16.99	100.00
Total	114,154	100.00	

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. tab state treat, m
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state of residence	SURE-P state (treated)		Total
	0	1	
abia	2,523	0	2,523
adamawa	3,223	0	3,223
akwa ibom	2,756	0	2,756
anambra	0	2,795	2,795
bauchi	0	3,533	3,533
bayelsa	2,840	0	2,840
benue	3,133	0	3,133
borno	3,041	0	3,041
cross river	2,248	0	2,248
delta	2,708	0	2,708
ebonyi	3,349	0	3,349
edo	2,583	0	2,583
ekiti	2,380	0	2,380
enugu	2,711	0	2,711
fct-abuja	2,799	0	2,799
gombe	3,437	0	3,437
imo	2,322	0	2,322
jigawa	3,635	0	3,635
kaduna	0	3,934	3,934
kano	5,448	0	5,448
katsina	3,980	0	3,980
kebbi	3,485	0	3,485
kogi	2,644	0	2,644
kwara	2,609	0	2,609
lagos	4,179	0	4,179
nasarawa	2,948	0	2,948
niger	0	3,283	3,283
ogun	2,112	0	2,112
ondo	0	2,561	2,561
osun	2,784	0	2,784
oyo	2,513	0	2,513
plateau	2,973	0	2,973
rivers	2,816	0	2,816
sokoto	3,324	0	3,324
taraba	3,875	0	3,875
yobe	3,377	0	3,377
zamfara	0	3,293	3,293
Total	94,755	19,399	114,154